<u>PROBLEM SCOPING</u>

❖ Sales Prediction System

Business Scenario

<u>Retail Company</u>

• Sales have become unpredictable in recent years.

• Some products sell very well, while others remain unsold.

• Overstocking and understocking are causing financial losses.

• Management wants to forecast future sales to improve inventory planning and business decisions.

They Approach an AI Team / Organization

• Asked to build an AI model/system to predict future sales.

• Help management optimize inventory and reduce business losses.

• Improve business planning and revenue generation.

Solution :

Understand the Business Problem

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Problem Statement

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Need of AI

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IDENTIFY STAKEHOLDERS

↓

Defining the Task of AI

↓

DEFINING SUCCESS METRICS

↓

DEFINING CONSTRAINTS

↓

ASSUMPTIONS AND RISKS

Step 1 : Understand the Business Problem

Initial Statement:

"Company sales fluctuate every month, making inventory planning difficult."

Need to uncover the actual problem and define measurable objectives.

Questions for Stakeholders

 # To Sales Department

• What are the current monthly sales?

• Which products have the highest sales?

• Which products have the lowest sales?

• During which months are sales highest?

 # To Inventory Department

• Which products frequently go out of stock?

• Which products remain unsold?

• How often do inventory shortages occur?

 # To Marketing Department

• Which marketing campaigns improve sales?

• Do discounts increase customer demand?

• Which advertisements perform best?

 # To Management

• What is the target sales growth?

• What budget is available?

• How will the success of the AI system be measured?

Suppose the Company Provides Following Reports

• Current yearly sales growth: 8%

• Desired yearly sales growth: 20%

• Sales increase during festivals.

• Discounts significantly improve demand.

• Stock shortages reduce revenue.

• Better forecasting can reduce inventory costs.

• Seasonal demand affects product sales.

Step 2 : Problem Statement

Predict future product sales before the next sales period so that management can optimize inventory, production, and marketing.

➤ **What?**

Predict future product sales.

➤ **When?**

Before the next sales cycle.

➤ **Why?**

Enable better inventory planning and business decisions.

Outcome:

Increase revenue and reduce inventory losses.

Step 3 : Need of AI

Can the problem be solved without AI?

If previous month's sales were high,

assume next month's sales will also be high.

Why AI is needed?

Sales depend on many factors:

1. Product Price
2. Discounts
3. Marketing Campaigns
4. Season
5. Festival Sales
6. Customer Demand
7. Product Ratings
8. Inventory Availability
9. Historical Sales
10. Weather Conditions

- Patterns are complex.
- Multiple factors interact.
- Manual forecasting is insufficient.

WITHOUT AI

Historical Sales

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Manual Forecast

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Stock Shortage / Overstock

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Business Loss

WITH AI

Historical Sales Data

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Sales Prediction Model

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Future Sales Forecast

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Inventory Planning

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Higher Revenue

STEP 4 : IDENTIFY STAKEHOLDERS

• **Company Management**

Improve revenue and profit.

• **Sales Team**

Improve sales planning.

• **Inventory Team**

Maintain optimal stock.

• **Marketing Team**

Plan better promotional campaigns.

• **Customers**

Receive products on time.

• **IT Department**

Deploy and maintain the AI system.

Step 5 : Defining the Task of AI

Machine Learning Formulation

Regression Problem

Output:

Future Sales Value

Example:

- Product A = 1200 Units
- Product B = 850 Units
- Product C = 500 Units

Forecasting Problem

Predict sales for:

- Next Week
- Next Month
- Next Quarter

Ranking Problem

Which products should receive higher inventory priority?

✓ Rank 1 = Highest Demand

✓ Rank 2 = Medium Demand

✓ Rank 3 = Low Demand

Step 6 : DEFINING SUCCESS METRICS

Business Metrics

• Increase yearly sales growth from 8% to 20%.

• Reduce inventory losses.

• Improve demand forecasting.

• Increase customer satisfaction.

Technical Metrics

• Mean Absolute Error (MAE)

• Root Mean Square Error (RMSE)

• R^2 Score

• Mean Absolute Percentage Error (MAPE)

Target:

• R^2 Score > 90%

• Low MAE

• Low RMSE

Step 7 : DEFINING CONSTRAINTS

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CONSTRAINTS

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DATA OPERATIONAL TIME ETHICAL

Operational Constraints

- Limited warehouse capacity.
- Limited inventory budget.

Time Constraints

- Predictions must be available before inventory planning.

Data Constraints

- Missing historical sales.
- Missing promotion records.
- Incorrect inventory data.
- Seasonal demand information may be unavailable.

Ethical Constraints

- Customer data privacy must be protected.
- Predictions should support decisions, not replace them.
- AI should not create unfair business decisions.

Step 8 : ASSUMPTIONS AND RISKS

Assumptions

- Historical sales reflect future demand.
- Sales records are accurate.
- Marketing campaigns influence customer demand.

Risks

- Missing data.
- Economic slowdown.
- Seasonal demand fluctuations.
- New competitors entering the market.
- Sudden changes in customer preferences.

Statistical Approach

Historical Sales Data

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Descriptive Statistics

(Mean, Median, Trend)

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Sales Trend Analysis

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Feature Selection

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Regression Model

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Future Sales Prediction